Introduction to Multimodal AI for Biomedical Applications

[W5-1: Neural Network – Training (2)]

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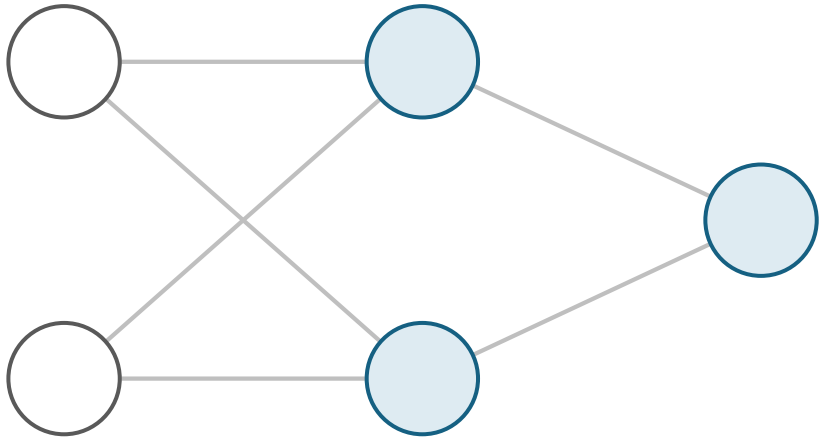
Department of AI

Today’s Journey

1	**Where the cost comes from**	work per slope × repeats
2	**Slopes from formulas**	derivatives, partial derivatives
3	**Gradient descent, exactly**	a small model, worked by hand
4	**Slopes through many steps**	the chain rule
5	**One slope in the network**	three steps, multiplied

By the end of this lecture, we will calculate one slope exactly.

The Network and One Example

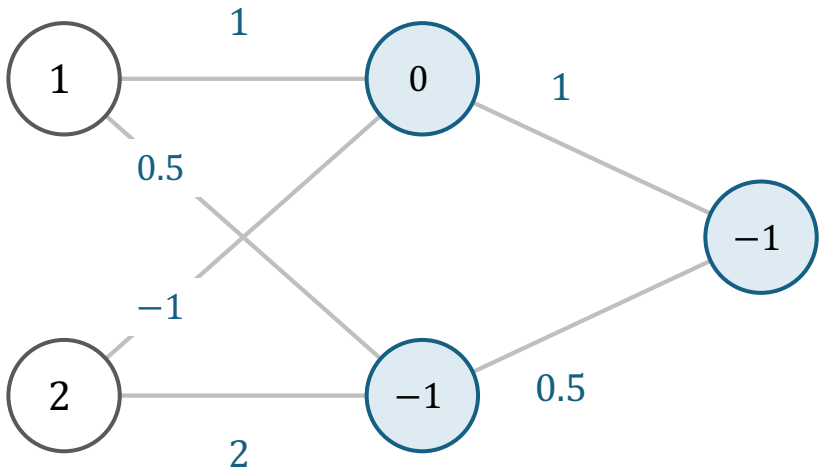


2 inputs

2 middle neurons (ReLU)

1 output neuron (sigmoid)

The Network and One Example



6 weights on the arrows

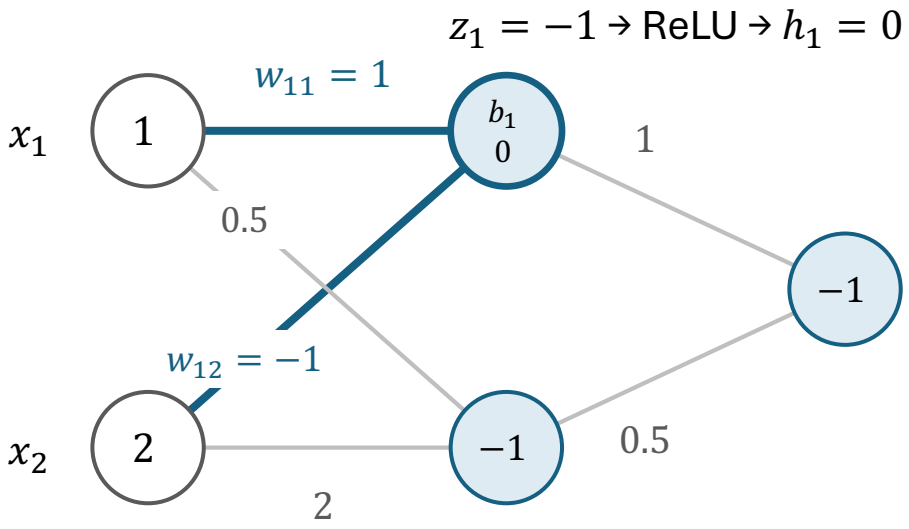
3 biases inside the neurons

nine numbers in total

one example

inputs 1 and 2, label $y = 1$

The Network and One Example



middle neuron 1

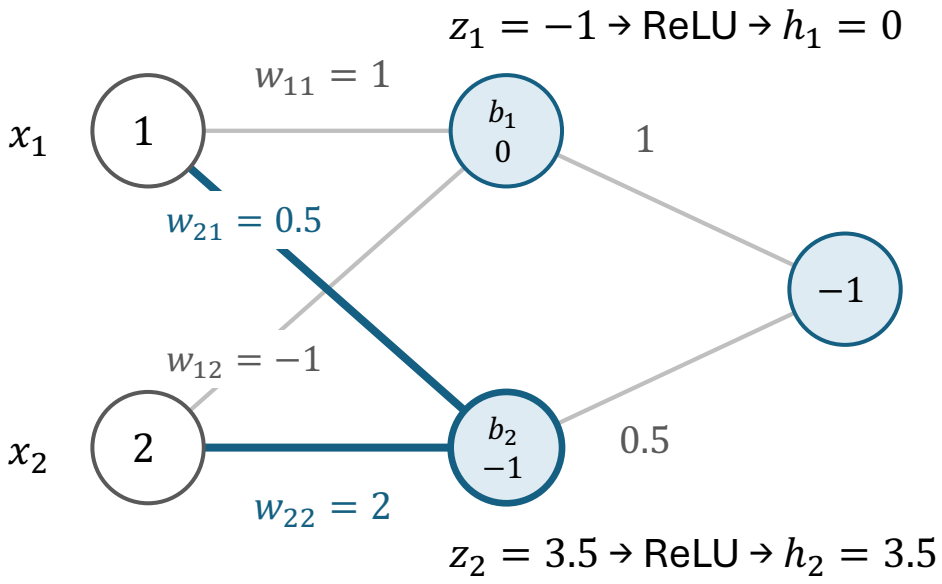
$$z_1 = w_{11}x_1 + w_{12}x_2 + b_1$$
$$= 1\times1 + (-1)\times2 + 0 = -1$$
$$\mathbf{h_1 = ReLU(z_1) = 0}$$

w_{ij}: from input j to neuron i

weighted sum: each input × the weight on its arrow, plus the bias

ReLU: negative → 0, positive → unchanged

The Network and One Example

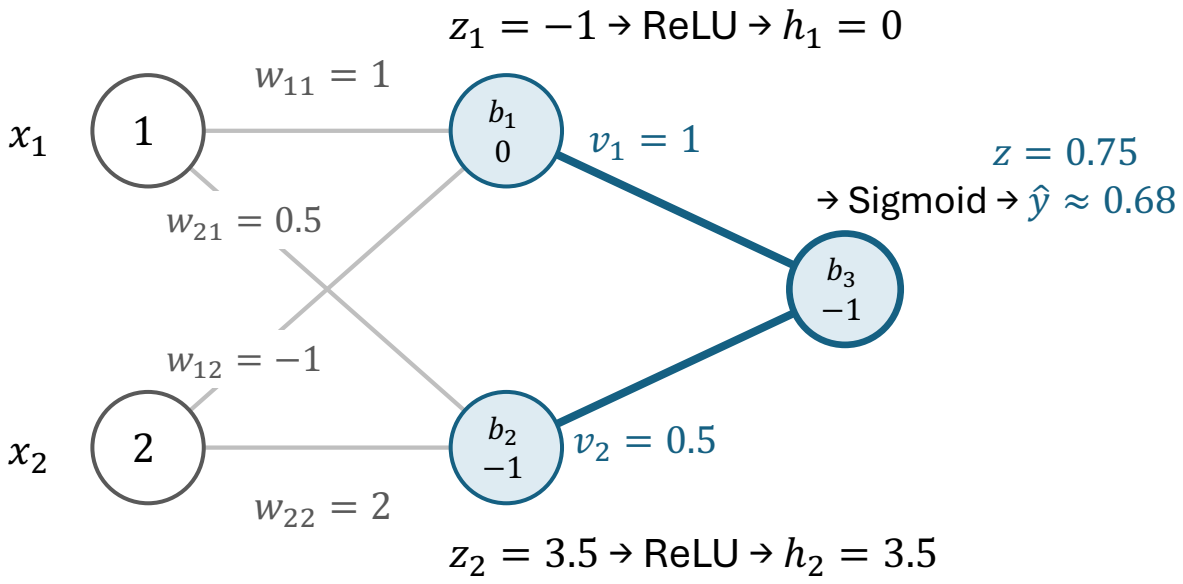


middle neuron 2

$$z_2 = w_{21}x_1 + w_{22}x_2 + b_2$$
$$= 0.5\times1 + 2\times2 + (-1) = 3.5$$
$$\mathbf{h_2 = ReLU(z_2) = 3.5}$$

same rule, with its own weights and bias

The Network and One Example



v_1, v_2: weights into the output neuron

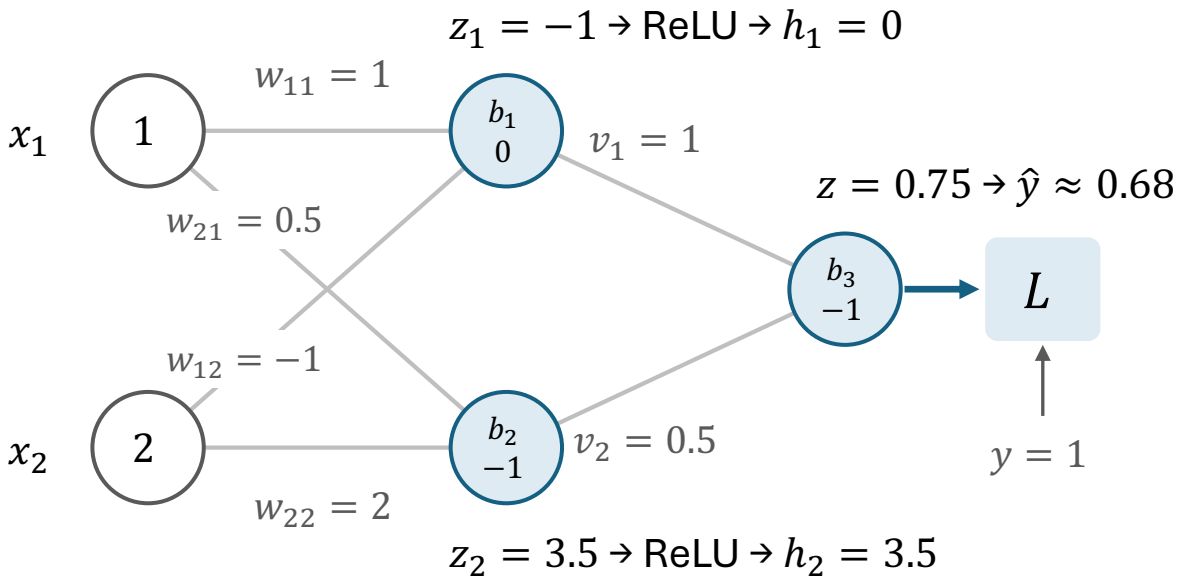
output neuron

$$z = v_1 h_1 + v_2 h_2 + b_3$$
$$= 1 \times 0 + 0.5 \times 3.5 + (-1) = 0.75$$
$$\hat{\mathbf{y}} = \boldsymbol{\sigma}(\mathbf{z}) \approx \mathbf{0.68}$$

sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The Network and One Example



the loss

$$L = (y - \hat{y})^2 = (1 - 0.68)^2$$
$$\approx \mathbf{0.1029}$$

inputs → loss: a **forward pass**

One forward pass turns the inputs and the nine numbers into a loss.

Every Weight Needs a Slope

nudge one weight

weight $0.5 \rightarrow 0.51$

loss $0.1029 \rightarrow 0.0981$

$$\frac{-0.0048}{0.01} = -0.48$$

the slope for this weight

the update rule

$$w_{\text{new}} = w_{\text{old}} - \eta \times \text{slope}$$

η: the learning rate

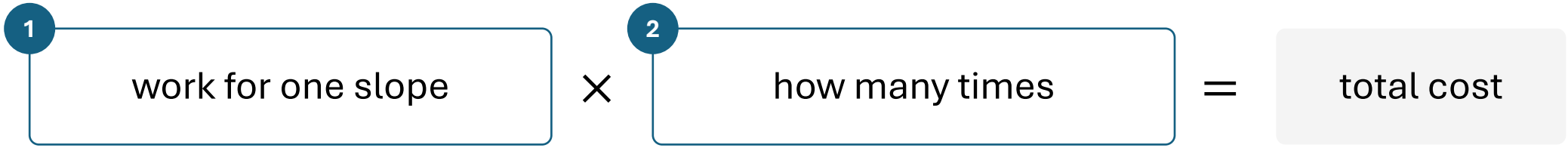
always move against the slope

the gradient = the list of all nine slopes

Training needs the slope of the loss for every one of the nine numbers.

Where the Cost Really Comes From

nudging: change one weight, run the whole network, compare the loss
repeated for every weight: $1 + N$ runs (N: number of weights)



nudging: a full run, and only an estimate

this lecture:
calculate it exactly, with derivatives

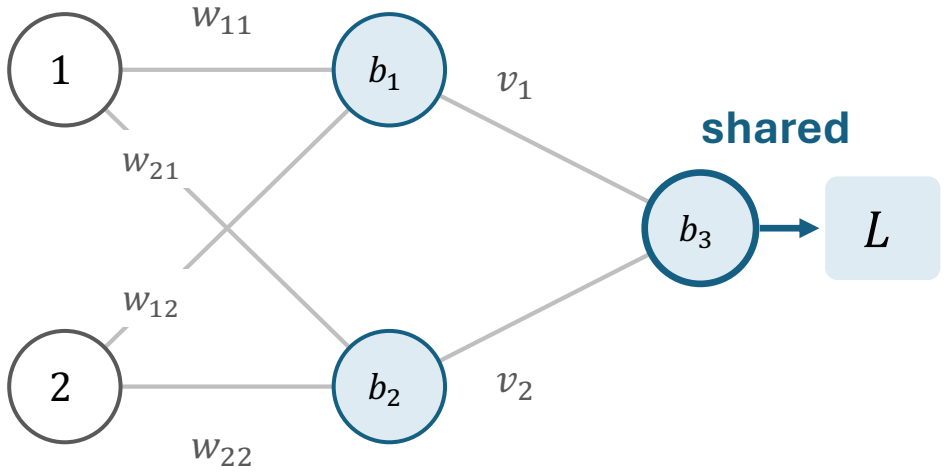
once per weight: N times

next lecture:
share the work, with backpropagation

② is what grows with the size of the network

The cost is the work for one slope, times how often it is repeated.

Every Path Ends the Same Way



A change in any of the nine numbers passes through the output neuron, then $\hat{y}$, then L.

Handled one weight at a time, that shared final stretch is worked through **nine times**.

Deeper networks share far more of every path.

Can the shared part be done once?

Handled one weight at a time, the shared final stretch is repeated.

From Nudging to the Derivative

$$f(w) = w^2, \text{ at } w = 3\text{: } f = 9$$

$$\frac{(3 + 0.1)^2 - 9}{0.1} = 6.1$$

nudge	ratio
0.1	6.1
0.01	6.01
0.001	6.001
→ 0	**6**

average rate of change

the nudge ratio: an average slope

↓ shrink the nudge

instantaneous rate of change

= the derivative: the exact slope at that point

written $f'(w)$ or $\frac{df}{dw}$

w^2 at $w = 3$: 6

Shrink the nudge: the average rate of change becomes the derivative.

Four Rules for Derivatives

	formula	derivative	example
1	a constant	0	$36 \to 0$
2	w^n	nw^{n-1}	$w^2 \to 2w, \;\; w \to 1$
3	a number × a formula	the number × its derivative	$5w^2 \to 10w$
4	a sum	the sum of the derivatives	term by term

$$4w^2 - 24w + 36 \quad \to \quad \mathbf{8w - 24}$$

Four rules give the derivative of any polynomial directly.

One Variable at a Time

$f(x, y) = x^2 y$ at $x = 3,\ y = 2$: $f = 18$

fix $y = 2$

$f = x^2 \times 2 = 2x^2$

a function of x alone

$\frac{d}{dx}(2x^2) = 4x$

at $x = 3$: **12**

fix $x = 3$

$f = 3^2 \times y = 9y$

a function of y alone

$\mathbf{\frac{d}{dy}(9y) = 9}$

One Variable at a Time

$$f(x, y) = x^2 y \quad \text{at } x = 3,\ y = 2: \quad f = 18$$

treat y as a constant

$$\frac{\partial f}{\partial x} = 2xy$$

at (3, 2): 12

treat x as a constant

$$\frac{\partial f}{\partial y} = x^2$$

at (3, 2): 9

partial derivative: change one variable, hold all the others fixed

∂ ("partial") instead of d: the other variables are held fixed

the slope of the loss for one weight is a partial derivative

A partial derivative changes one variable and holds all the others fixed.

Derivatives of a Weighted Sum

$$z = w_1 x_1 + w_2 x_2 + b$$

x_1, x_2: inputs w_1, w_2: their weights b: bias z: weighted sum

$$\frac{\partial z}{\partial w_1} = x_1$$

a weight's slope is its input

$$\frac{\partial z}{\partial x_1} = w_1$$

an input's slope is its weight

$$\frac{\partial z}{\partial b} = 1$$

the bias is simply added

an input of 0 → that weight has no influence on z

A weight's partial derivative is its input; an input's is its weight.

The Toy Model's Exact Slope

input: 2
prediction: $\hat{y} = 2w$
label: 6

$$L = (6 - 2w)^2$$
$$= 36 - 24w + 4w^2$$
$$\frac{\partial L}{\partial w} = 8w - 24$$

at $w = 1$

$$\frac{\partial L}{\partial w} = 8 \times 1 - 24 = -16$$

negative slope: increasing w lowers the loss

From the formula, the slope at the start is minus sixteen.

Gradient Descent, Exactly

$$\hat{y} = 2w,\ L = (6 - \hat{y})^2,\quad \frac{\partial L}{\partial w} = 8w - 24,\quad w_{\text{new}} = w_{old} - \eta \times \frac{\partial L}{\partial w},\quad \eta = 0.025$$

step 0: start at w = 1

1	slope	$\frac{\partial L}{\partial w} = 8 \times 1 - 24 = -16$
2	update	$w_{\text{new}} = 1 - 0.025 \times (-16) = 1 + 0.4 = 1.4$
3	new loss	$\hat{y} = 2 \times 1.4 = 2.8,\quad L = (6 - 2.8)^2 = 10.24$

Gradient Descent, Exactly

$$\hat{y} = 2w,\ L = (6 - \hat{y})^2,\quad \frac{\partial L}{\partial w} = 8w - 24,\quad w_{\text{new}} = w_{old} - \eta \times \frac{\partial L}{\partial w},\quad \eta = 0.025$$

step 1: start at w = 1.4

1	slope	$\frac{\partial L}{\partial w} = 8 \times 1.4 - 24 = -12.8$
2	update	$w_{\text{new}} = 1.4 - 0.025 \times (-12.8) = 1.4 + 0.32 = 1.72$
3	new loss	$\hat{y} = 2 \times 1.72 = 3.44,\quad L = (6 - 3.44)^2 \approx 6.55$

Gradient Descent, Exactly

$$\hat{y} = 2w,\ L = (6 - \hat{y})^2,\quad \frac{\partial L}{\partial w} = 8w - 24,\quad w_{\text{new}} = w_{old} - \eta \times \frac{\partial L}{\partial w},\quad \eta = 0.025$$

step	w	① slope	② new w	③ new loss
0	1	−16	1.4	10.24
1	1.4	−12.8	1.72	6.55
2	1.72	−10.24	1.976	4.19
3	1.976	−8.19	2.18	2.68

starting loss: 16

the model is right at $w = 3$

Each iteration: slope from the formula, update, repeat.

A Formula Too Long to Expand

1. $z_1 = w_{11}x_1 + w_{12}x_2 + b_1, \quad h_1 = \mathrm{ReLU}(z_1)$ — middle neuron 1
2. $z_2 = w_{21}x_1 + w_{22}x_2 + b_2, \quad h_2 = \mathrm{ReLU}(z_2)$ — middle neuron 2
3. $z = v_1h_1 + v_2h_2 + b_3$ — output: weighted sum
4. $\hat{y} = \sigma(z)$ — output: sigmoid
5. $L = (y - \hat{y})^2$ — loss

all five steps in one line: **cannot be multiplied out**

$$L = \left(y - \sigma(v_1\mathrm{ReLU}(w_{11}x_1 + w_{12}x_2 + b_1) + v_2\mathrm{ReLU}(w_{21}x_1 + w_{22}x_2 + b_2) + b_3)\right)^2$$

Can we find the derivative one step at a time?

A network's loss is a formula nested many steps deep.

A Change Travels in Steps

$w = 1$ → $\hat{y} = 2$ → $L = 16$

step 1

$$\hat{y} = 2w$$

$$\frac{\partial \hat{y}}{\partial w} = 2$$

step 2

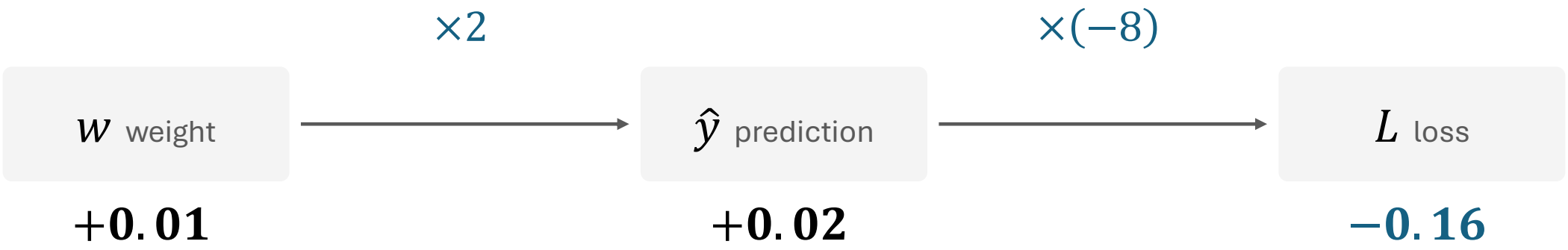
$$L = (6 - \hat{y})^2 = 36 - 12\hat{y} + \hat{y}^2$$

$$\frac{\partial L}{\partial \hat{y}} = -12 + 2\hat{y} = -8$$

How do the two derivatives combine?

A weight reaches the loss only through the steps in between.

Following One Small Change

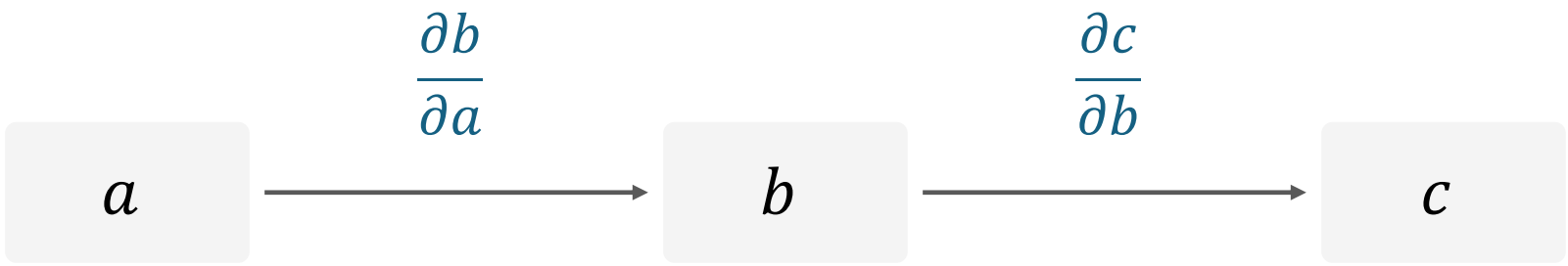


$$\frac{-0.16}{0.01} = -16 = 2\times(-8)$$

the same -16 as $8w - 24$ at $w = 1$

Each step multiplies the change that arrives by its own derivative.

The Chain Rule



the chain rule

$$\frac{\partial c}{\partial a} = \frac{\partial c}{\partial b} \times \frac{\partial b}{\partial a}$$

five steps → five derivatives multiplied

each factor at the current values

The chain rule: along a chain of steps, the derivatives multiply.

The Derivative of the Loss

$$L = (y - \hat{y})^2$$
$$= y^2 - 2y\hat{y} + \hat{y}^2$$

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y)$$

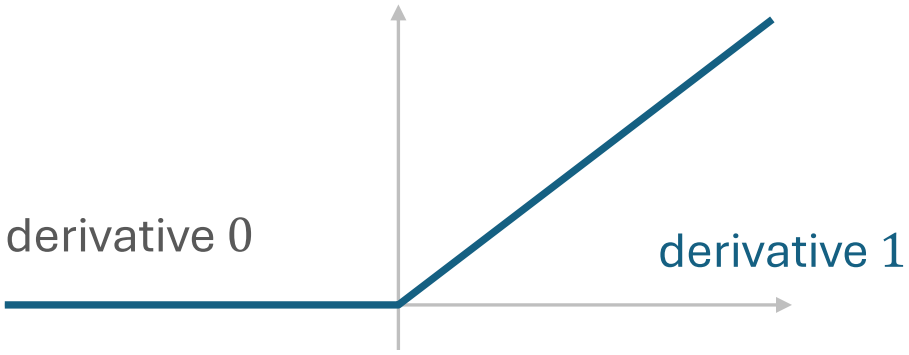
our network: $2 \times (0.68 - 1) \approx -0.64$

y: the label (correct answer) $\hat{y}$: the prediction

The derivative of a squared error is two times prediction minus label.

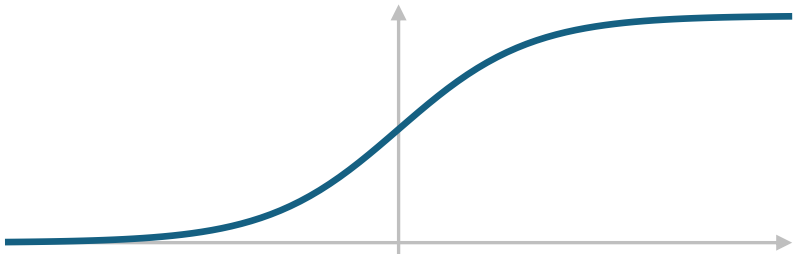
The Derivative of an Activation

ReLU



negative input → derivative 0
positive input → derivative 1

sigmoid

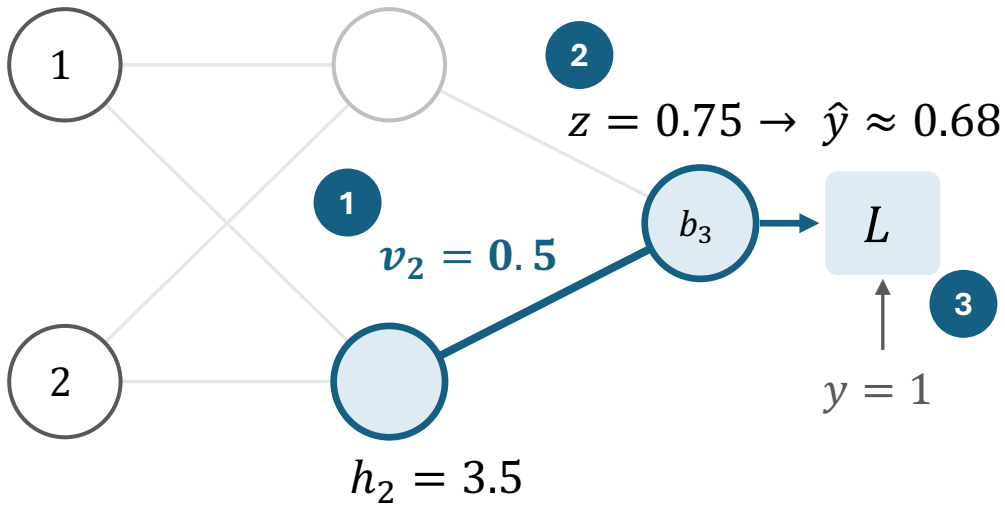


$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

$$\sigma(0) = \frac{1}{2} \quad \rightarrow \quad \sigma'(0) = \frac{1}{4}$$

An activation's derivative depends only on the value that passed through it.

One Weight's Path to the Loss



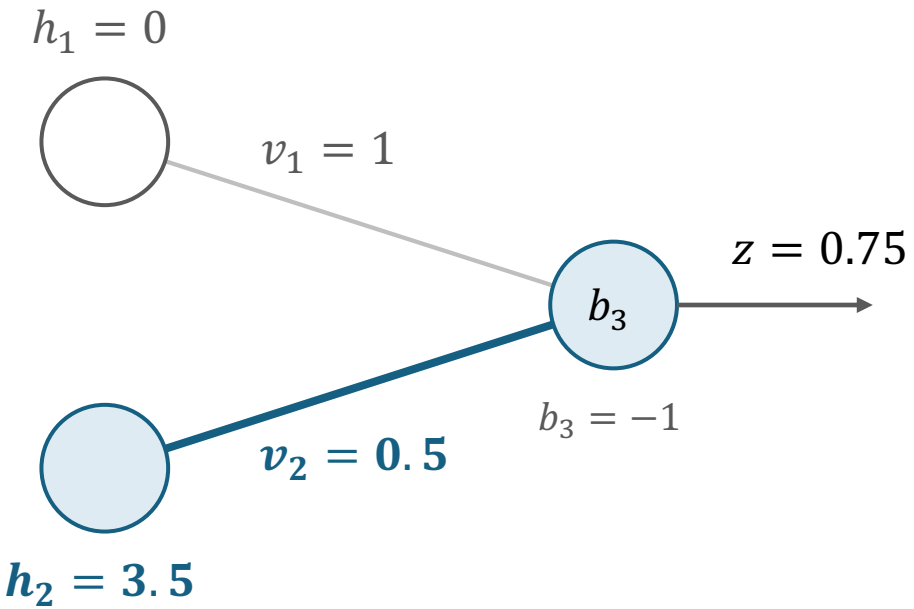
1. v_2 changes z, the output neuron's weighted sum
2. z goes through the sigmoid and changes $\hat{y}$
3. $\hat{y}$ changes the loss L

$$\frac{\partial L}{\partial v_2} = \overset{3}{\frac{\partial L}{\partial \hat{y}}} \times \overset{2}{\frac{\partial \hat{y}}{\partial z}} \times \overset{1}{\frac{\partial z}{\partial v_2}}$$

v_2: the weight on the arrow from middle neuron 2 to the output

The weight reaches the loss in three steps, each with a derivative.

Step 1: From the Weight to the Sum



$$z = v_1 h_1 + v_2 h_2 + b_3$$

only the coloured term contains the weight

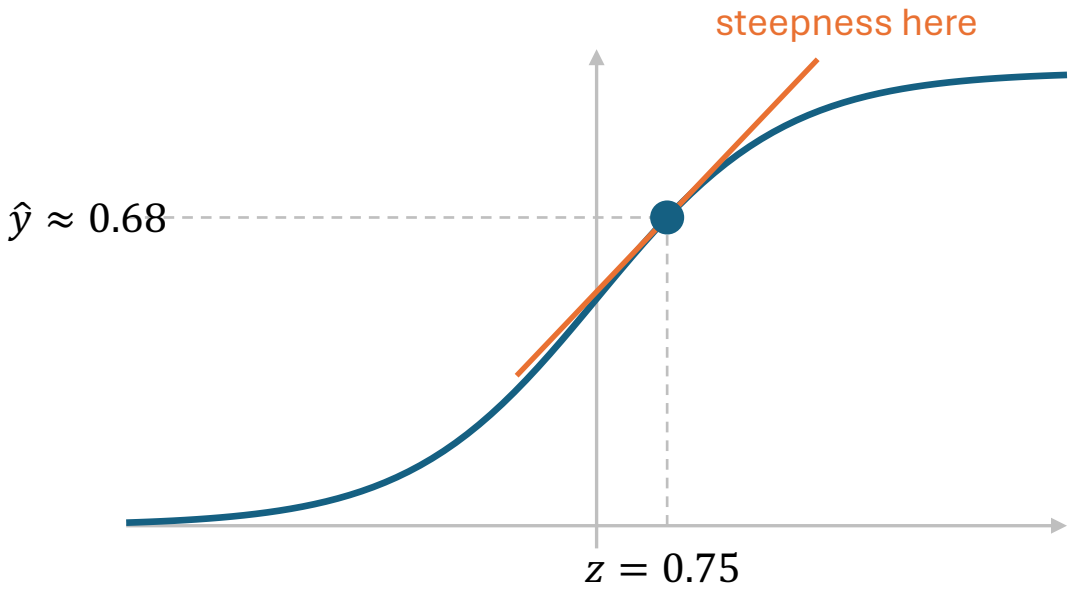
1

$$\frac{\partial z}{\partial v_2} = h_2 = 3.5$$

the value arriving on the same arrow as v_2

Step 1: the derivative is the value arriving on the same arrow.

Step 2: Through the Sigmoid



$$\hat{y} = \sigma(z)$$

2

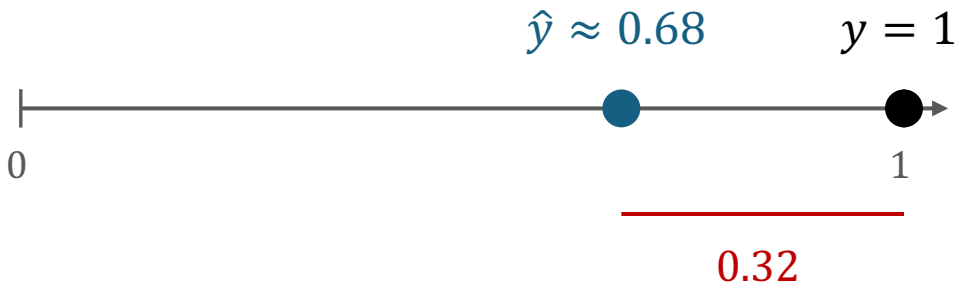
$$\frac{\partial \hat{y}}{\partial z} = \sigma'(z) = \hat{y}(1 - \hat{y})$$

$$= 0.68 \times (1 - 0.68)$$

$$= \mathbf{0.68 \times 0.32 \approx 0.22}$$

Step 2: the sigmoid's derivative uses only its own output.

Step 3: Into the Loss



the prediction is below the label

$$L = (y - \hat{y})^2$$

3

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y)$$

$$= 2\times(0.68 - 1)$$

$$= \mathbf{2\times(-0.32) \approx -0.64}$$

Step 3: the loss derivative is two times prediction minus label.

Multiplying the Three Steps

$$\frac{\partial L}{\partial v_2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial v_2}$$

	3	2	1
	$\frac{\partial L}{\partial \hat{y}}$	$\frac{\partial \hat{y}}{\partial z}$	$\frac{\partial z}{\partial v_2}$
	the slope we want	into the loss	through the sigmoid	value on the arrow, h_2
	≈ -0.49	-0.64	0.22	3.5

from the loss end: $-0.64 \times 0.22 \approx \boxed{-0.14}$, then $-0.14 \times 3.5 \approx -0.49$

calculated exactly, from values the forward pass already has

One slope, calculated exactly from values the forward pass already has.

Next: Sharing the Work

this lecture

cost = work for one slope × how many times

derivatives, partial derivatives, the chain rule

one slope, exactly: $\frac{\partial L}{\partial v_2} \approx -0.49$

next lecture

the work is still done once for every weight

their paths to the loss share a great deal

Can the shared part be done once?

One slope is calculated; next, all of them without repeating the work.